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a) $\nabla\nabla_{12}X_t = \nabla(X_t - X_{t-12})$
$= X_t - X_{t-1} - (X_{t-12} - X_{t-13})$
$= (\alpha + \beta t + s_t + Y_t)$
$- (\alpha + \beta(t-1) + s_{t-1} + Y_{t-1})$
$- (\alpha + \beta(t-12) + s_{t-12} + Y_{t-12})$
$+ (\alpha + \beta(t-13) + s_{t-13} + Y_{t-13})$
$= Y_t - Y_{t-1} - Y_{t-12} + Y_{t-13}$

$\nabla\nabla_{12}X_t$ is stationary b/c $E[\nabla\nabla_{12}X_t] = \overbrace{1-1-1+1}^{\text{coefficients of } Y_t} = 0$
and $Cov(\nabla\nabla_{12}X_{t+h}, \nabla\nabla_{12}X_t)$ = some combination of $\gamma_Y(h)$ which clearly doesn't depend on t.

ACVF
By property 4 of covariance:

$\gamma(h) = \sum_a \sum_b c_a c_b Cov(Y_{t+h-a}, Y_{t-b})$
$= \sum_a \sum_b c_a c_b \gamma_Y(h-a+b)$ by linearity of Cov., $\overbrace{(t+h-a)-(t-b)}^{\text{lag subscripts of } Y} = h-a+b$

For $a, b \in \{0, 1, 12, 13\}$ (the Y_t lags above),
and c_i is the coefficient (-1 or 1) for the corresponding Y_i for $i = a$ or b